

API Automation Testing Report

Project Name: Restful-booker.herokuapp API Testing with RestAssured

Participant: Shahbaz Ahmad

Date: 25-06-2026

Branch: shahbaz_api_25_june_2006(git)

Assessment: API Testing

Question NO 1

Objective

To design, develop, and execute API automation tests ensuring reliability, scalability, and correctness of REST APIs using a structured framework. My objective is to test the website

'https://www.makemytrip.com/' Api by using crud operation
GET,POST,PUT,DELETE

GithubLink:-

I have created a dedicated branch for this api testing assessment and branch name is

[ust-sdet-training/SDET at shahbaz 25 June 2026](#)

Scenario

Get,post and delete and put secure order creation and fetch flow with authentication token.

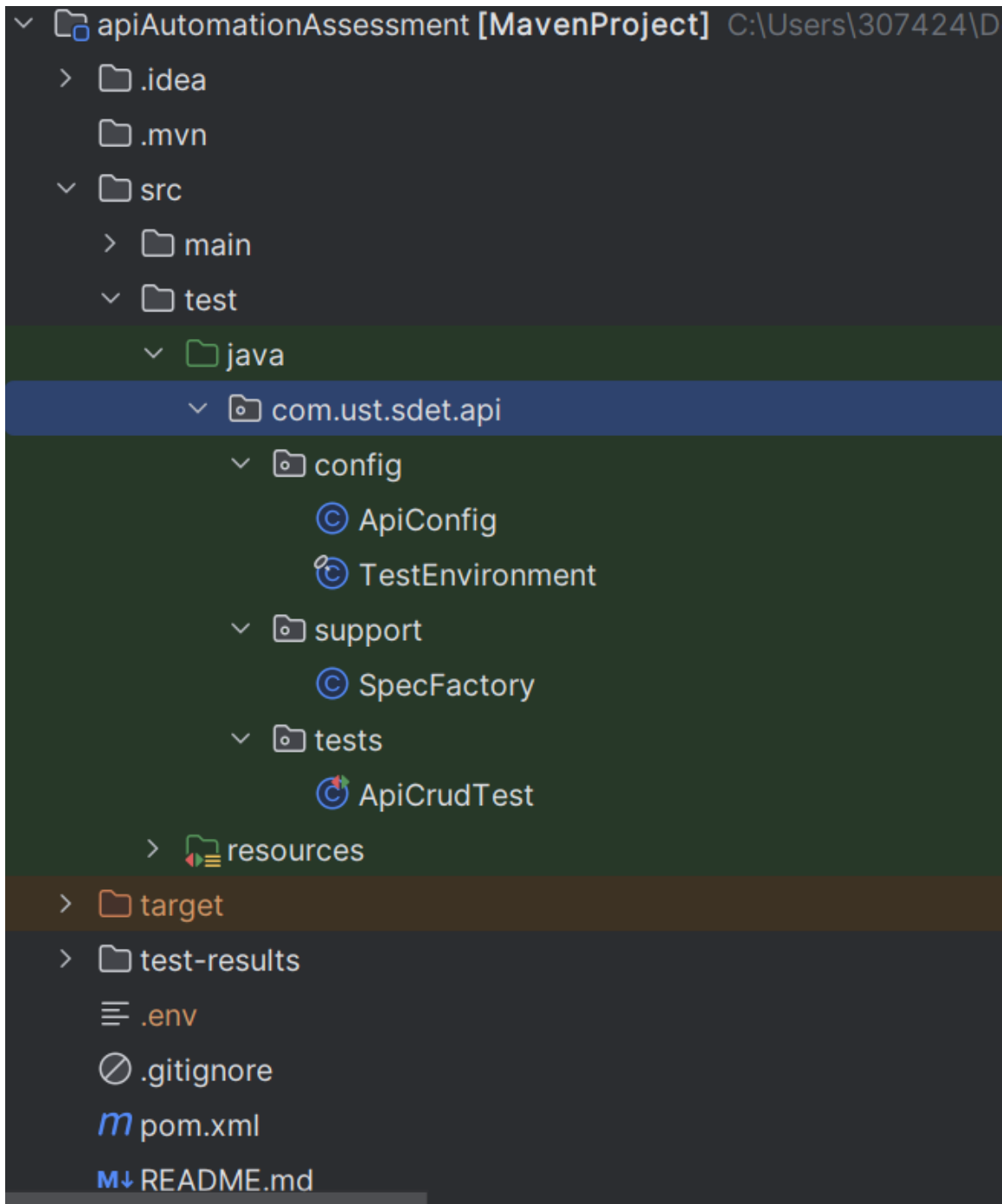
Tools & Technologies

- **Testing Framework:** JUnit
- **API Library:** RestAssured
- **Assertions:** Hamcrest
- **API Documentation:** Swagger UI
- **Design Pattern:** SpecFactory (Reusable Request/Response Specifications)

Environment Setup

Java, Maven, RestAssured, IntelliJ IDEA, Allure Reporting. Environment variables managed via .env and TestEnvironment utility class.

Folder Structure



Workflow

1. Read configuration from .env
2. Generate token

3. Build request using SpecFactory
4. Execute API
5. Validate response
6. Assertion using hamcrest,junit
7. Generate report

Explanation

1.TestEnvironmet

```
package com.ust.sdet.api.config;

import ...

public final class TestEnvironment { 3 usages @ Shahbaz Ahmad(UST
    private static final Path ENV_FILE = findEnvFile(); 3 usages
    private static final Map<String, String> LOCAL_ENV = loadLocalEnv(ENV_FILE); 3 usages

    private TestEnvironment() { no usages @ Shahbaz Ahmad(UST
    }

    public static String required(String name) { 1 usage @ Shahbaz Ahmad(UST
        String value = optional(name, fallback: null);
        if (value == null) {
            throw new IllegalStateException(
                "Missing " + name + ". Set it in the IntelliJ run configuration or in .env. "
                + "Working directory: " + Path.of(S. "").toAbsolutePath()
                + ". Loaded .env: " + loadedEnvFile()
            );
        }
        return value;
    }

    public static String optional(String name, String fallback) { 1 usage @ Shahbaz Ahmad(UST
        String systemValue = System.getProperty(name);
        if (systemValue != null && !systemValue.isBlank()) {
            return systemValue;
        }

        String environmentValue = System.getenv(name);
        if (environmentValue != null && !environmentValue.isBlank()) {
            return environmentValue;
        }

        String fileValue = LOCAL_ENV.get(name);
```

```

public static String optional(String name, String fallback) { 1 usage & Shahbaz Ahmad(UST
    String systemValue = System.getProperty(name);
    if (systemValue != null && !systemValue.isBlank()) {
        return systemValue;
    }

    String environmentValue = System.getenv(name);
    if (environmentValue != null && !environmentValue.isBlank()) {
        return environmentValue;
    }

    String fileValue = LOCAL_ENV.get(name);
    return fileValue == null || fileValue.isBlank() ? fallback : fileValue;
}

public static String loadedEnvFile() { return ENV_FILE == null ? "none" : ENV_FILE.toString(); }

public static String sourceOf(String name) { no usages & Shahbaz Ahmad(UST
    String systemValue = System.getProperty(name);
    if (systemValue != null && !systemValue.isBlank()) {
        return "JVM system property (-D" + name + ")";
    }
    String environmentValue = System.getenv(name);
    if (environmentValue != null && !environmentValue.isBlank()) {
        return "process environment / IntelliJ run configuration";
    }
    if (LOCAL_ENV.containsKey(name) && !LOCAL_ENV.get(name).isBlank()) {
        return ".env file: " + loadedEnvFile();
    }
    return "not configured";
}

```

```

private static Path findEnvFile() { 1 usage 2 Shahbaz Ahmad(UST
    String explicitPath = System.getProperty("envFile");
    if (explicitPath == null || explicitPath.isBlank()) {
        explicitPath = System.getenv( s: "ENV_FILE");
    }
    if (explicitPath != null && !explicitPath.isBlank()) {
        Path configured = Path.of(explicitPath).toAbsolutePath().normalize();
        if (!Files.isRegularFile(configured)) {
            throw new IllegalStateException("Configured ENV_FILE does not exist: " + configured);
        }
        return configured;
    }

    Set<Path> startingPoints = new LinkedHashSet<>();
    startingPoints.add(Path.of( s: "").toAbsolutePath().normalize());
    try {
        Path codeLocation = Path.of(
            TestEnvironment.class.getProtectionDomain().getCodeSource().getLocation().toURI()
        ).toAbsolutePath().normalize();
        startingPoints.add(codeLocation);
    } catch (URISyntaxException ignored) {
        // The working-directory search still provides a normal fallback.
    }

    for (Path startingPoint : startingPoints) {
        Path found = findUpwards(startingPoint);
        if (found != null) {
            return found;
        }
    }

    return null;
}

```

```

private static Path findUpwards(Path startingPoint) { 1 usage & Shahbaz Ahmad(UST
    Path current = Files.isDirectory(startingPoint) ? startingPoint : startingPoint.getParent();
    while (current != null) {
        Path candidate = current.resolve(s: ".env");
        if (Files.isRegularFile(candidate)) {
            return candidate;
        }
        current = current.getParent();
    }
    return null;
}

private static Map<String, String> loadLocalEnv(Path envFile) { 1 usage & Shahbaz Ahmad(UST
    if (envFile == null) {
        return Map.of();
    }

    Map<String, String> values = new HashMap<>();
    try {
        for (String line : Files.readAllLines(envFile)) {
            String trimmed = line.trim();
            if (trimmed.isEmpty() || trimmed.startsWith("#") || !trimmed.contains("=")) {
                continue;
            }
            if (trimmed.startsWith("export ")) {
                trimmed = trimmed.substring("export ".length()).trim();
            }
            String[] entry = trimmed.split(s: "=", t: 2);
            values.put(entry[0].trim(), unquote(entry[1].trim()));
        }
    } catch (IOException error) {

```

```

        continue;
    }
    if (trimmed.startsWith("export ")) {
        trimmed = trimmed.substring("export ".length()).trim();
    }
    String[] entry = trimmed.split(s: "=", k: 2);
    values.put(entry[0].trim(), unquote(entry[1].trim()));
}
} catch (IOException error) {
    throw new IllegalStateException("Unable to read .env file: " + envFile, error);
}
return values;
}

private static String unquote(String value) { 1 usage @ Shahbaz Ahmad(UST
    if (value.length() >= 2) {
        char first = value.charAt(0);
        char last = value.charAt(value.length() - 1);
        if ((first == '"' && last == '"') || (first == '\'' && last == '\'')) {
            return value.substring(i: 1, fi: value.length() - 1);
        }
    }
    return value;
}
}

```

1. required(String name)

- Retrieves a mandatory configuration value from available sources.
 - Internally calls optional() to fetch the value.
 - Throws an exception if the value is missing or blank.
 - Provides a detailed error message with working directory and .env file info.
 - Ensures critical environment variables are always present before test execution.
-

2. optional(String name, String fallback)

- Fetches configuration values from multiple sources in priority order.
 - First checks JVM system properties (-D arguments).
 - Then checks OS environment variables or IntelliJ run configuration.
 - Finally, falls back to values loaded from the .env file.
 - Returns a default fallback value if none are found.
-

3. loadedEnvFile()

- Returns the path of the .env file that was loaded.
 - If no file is found, it returns "none".
 - Helps in debugging configuration issues during execution.
 - Used in error messages to provide context.
 - Improves transparency of environment setup.
-

4. sourceOf(String name)

- Identifies the source of a given configuration variable.
- Checks whether the value came from JVM arguments, environment variables, or .env.
- Returns a human-readable source description.
- Helps debug which configuration override is being applied.
- Useful during test troubleshooting and environment validation.

5. findEnvFile()

- Locates the .env file dynamically.
- First checks explicitly configured path via system property or ENV variable.
- Then searches from working directory and code location.
- Uses upward directory traversal to locate the nearest .env.
- Ensures flexible configuration regardless of project structure.

6. findUpwards(Path startingPoint)

- Searches parent directories for a .env file.
- Starts from a given directory and moves upwards recursively.
- Stops when the file is found or root directory is reached.
- Returns the path of the .env file if found.
- Enables automatic environment file discovery.

7. loadLocalEnv(Path envFile)

- Reads and parses key-value pairs from the .env file.
 - Ignores comments (#) and empty lines.
 - Supports export syntax used in shell-style env files.
 - Stores variables in a Map for quick lookup.
 - Throws an exception if the file cannot be read.
-

8. unquote(String value)

- Removes surrounding quotes from environment variable values.
 - Supports both single (') and double (") quotes.
 - Ensures clean values without extra characters.
 - Prevents formatting issues during variable usage.
 - Helps maintain compatibility with .env file formats.
-

9. Static Initialization (ENV_FILE & LOCAL_ENV)

- Automatically executes when the class is loaded.
- Identifies the .env file using findEnvFile().
- Loads environment variables into memory via loadLocalEnv().
- Ensures configuration is ready before any test runs.
- Centralizes environment management for the entire framework.

2.ApiConfig

ApiConfig Class

- Centralizes API configuration values like URL, path, and test data.
 - Uses TestEnvironment.required() to ensure all values are present.
 - Provides globally accessible static variables for tests.
-

BASE_PATH

- Stores the base path for API endpoints.
- Loaded from environment configuration.

- Avoids hardcoding paths in test cases.
-

BASE_URL

- Represents the main API server URL.
- Retrieved dynamically from environment variables.
- Supports switching between environments easily.

3.SpecFactory

```
1 package com.ust.sdet.api.support;
2
3 import com.ust.sdet.api.config.TestEnvironment;
4 import io.restassured.builder.RequestSpecBuilder;
5 import io.restassured.builder.ResponseSpecBuilder;
6 import io.restassured.http.ContentType;
7 import io.restassured.specification.RequestSpecification;
8 import io.restassured.specification.ResponseSpecification;
9
10 import java.util.Map;
11 import static com.ust.sdet.api.config.ApiConfig.BASE_URL;
12 import static io.restassured.RestAssured.given;
13
14 public class SpecFactory { 1 usage 2 Shahbaz Ahmad(UST
15
16
17     public static RequestSpecification authedTokenCreatedRequest = 1 usage
18         new RequestSpecBuilder()
19             .setBaseUri(BASE_URL)
20             .setBasePath("")
21             .addHeader(headerName: "Content-Type", headerValue: "application/json")
22             .build();
23
24 }
```

```

public static RequestSpecification commonJsonRequest = 8 usages
    new RequestSpecBuilder()
        .setBaseUri(BASE_URL)
        .setBasePath("")
        .addHeader( headerName: "Content-Type", headerValue: "application/json")
        .build();

public static ResponseSpecification commonJsonResponse = 7 usages
    new ResponseSpecBuilder()
        .expectContentType(ContentType.JSON)
        .expectStatusCode( expectedStatusCode: 200)
        .build();

//token

public static RequestSpecification oauthTokenGenerateSpecRequest= new RequestSpecBuilder() no usages
    .setBaseUri(BASE_URL)
    .setBasePath("")
    .setContentType("application/x-www-form-urlencoded; charset=UTF-8")

```

```

//token

public static RequestSpecification oauthTokenGenerateSpecRequest= new RequestSpecBuilder() no usages
    .setBaseUri(BASE_URL)
    .setBasePath("")
    .setContentType("application/x-www-form-urlencoded; charset=UTF-8")
//    .addFormParam("grant_type",generateTokenBody.get("grant_type"))
    .build();
}

```

1. *authTokenCreatedRequest*

(RequestSpecBuilder)

- This builder creates a reusable request specification for order-related API calls.
- It sets the **base URI** and **base path** dynamically from environment variables.

- Adds a default header Content-Type: application/json to every request.
 - Helps avoid duplication of configuration across multiple API tests.
 - Intended for authenticated requests where headers/body will be extended later.
-

2. commonJsonRequest (RequestSpecBuilder)

- Defines a generic request specification for JSON-based API calls.
 - Configures the **base URI** and **base path** centrally for all requests.
 - Sets the Content-Type header to application/json by default.
 - Promotes consistency and reduces boilerplate code in test scripts.
 - Can be reused across multiple test cases requiring JSON payloads.
-

3. commonJsonResponse (ResponseSpecBuilder)

- Defines a standard response expectation for successful API calls.
 - Verifies that the response content type is JSON.
 - Ensures the HTTP status code is **200 (OK)**.
 - Reusable validation for most successful GET/POST requests.
 - Helps enforce uniform response validation across tests.
-

4. commonJsonNotFoundResponse (ResponseSpecBuilder)

- Handles scenarios where a requested resource is not found.

- Validates HTTP status code **404 (Not Found)**.
- Typically used in negative test cases.
- Ensures API correctly responds to invalid resource requests.
- Helps verify error handling and edge cases.

4.ApiCrudTest

```
1 package com.ust.sdet.api.tests;
2
3 import com.ust.sdet.api.config.ApiConfig;
4 import io.restassured.http.ContentType;
5 import io.restassured.path.json.JsonPath;
6 import io.restassured.response.Response;
7 import org.junit.jupiter.api.*;
8
9 import java.util.HashMap;
10 import java.util.List;
11 import java.util.Map;
12
13 import static com.ust.sdet.api.support.SpecFactory.*;
14 import static io.restassured.RestAssured.given;
15 import static org.hamcrest.Matchers.*;
16 import static org.hamcrest.MatcherAssert.assertThat;
17 import static org.junit.jupiter.api.Assertions.*;
18
19 @TestMethodOrder(MethodOrderer.OrderAnnotation.class) & Shahbaz Ahmad(UST
20 public class ApiCrudTest {
21
22     static String token; no usages
23     static int createdPetId; 1 usage
24     static Map<String, Object> requestBody; 4 usages
25
26     @BeforeAll & Shahbaz Ahmad(UST
27     static void setup() {
28         requestBody = Map.of(
29             k: "firstname", v: "Jim",
30             k1: "lastname", v1: "Brown",
31             k2: "totalprice", v2: 111,
32             k3: "depositpaid", v3: true,
33             k4: "bookingdates", Map.of(
34                 k: "checkin", v: "2018-01-01",
35                 k1: "checkout", v1: "2019-01-01"
```

@AfterAll & Shahbaz Ahmad(UST

static void tearDown() { System.out.println("Test execution completed "); }

public String testCreateToken() { 2 usages & Shahbaz Ahmad(UST

```
var requestBody : Map<String, String> = Map.of( k: "username", v: "admin", k1: "password", v1: "password123");
var response : Response = given() RequestSpecification
    .spec(authedTokenCreatedRequest)
    .body(requestBody)
    .when()
    .post( path: "/auth") Response
    .then() ValidatableResponse
    .spec(commonJsonResponse)
    .extract().response();
```

```
JsonPath json = response.jsonPath();
return json.getString( path: "token");
```

}

@Test & Shahbaz Ahmad(UST

@Order(1)

@DisplayName("GET - get all booking")

void testGetAllBooking() {

```
var response : Response = given() RequestSpecification
    .spec(commonJsonRequest)
    .when()
    .get( path: "/booking") Response
    .then() ValidatableResponse
    .spec(commonJsonResponse)
    .extract() ExtractableResponse<Response>
    .response();
```

```
    assertNotNull(response.jsonPath().getInt( path: "[0].bookingid"));
}
```

```
@Test & Shahbaz Ahmad(UST
```

```
@Order(2)
```

```
@DisplayName("Post - Create new booking")
```

```
void createBooking() {
```

```
    var response : Response = given() RequestSpecification
```

```
        .spec(commonJsonRequest)
```

```
        .body(requestBody)
```

```
        .when()
```

```
        .post( path: "/booking") Response
```

```
        .then() ValidatableResponse
```

```
        .spec(commonJsonResponse)
```

```
        .extract() ExtractableResponse<Response>
```

```
        .response();
```

```
    JsonPath json = response.jsonPath();
```

```
    createdPetId = json.getInt( path: "bookingid");
```

```
    assertTrue( condition: json.getInt( path: "bookingid")>=0);
```

```
}
```

```
@Test & Shahbaz Ahmad(UST
```

```
@Order(3)
```

```
@DisplayName("GET - Fetch booking By ID")
```

```
void getBookingById() {
```

```
    var response : Response = given() RequestSpecification
```

```
        .spec(commonJsonRequest)
```

```

5  @Test & Shahbaz Ahmad(UST
6  @Order(3)
7  @DisplayName("GET - Fetch booking By ID")
8  void getBookingById() {
9
10     var response :Response = given() RequestSpecification
11         .spec(commonJsonRequest)
12         .pathParam( parameterName: "id", parameterValue: 2)
13         .when()
14         .get( path: "/booking/{id}") Response
15         .then() ValidatableResponse
16         .spec(commonJsonResponse)
17         .extract() ExtractableResponse<Response>
18         .response();
19
20     //
21     assertEquals(createdPetId,response.jsonPath().getInt("bookingid"));
22     assertNotNull(response.jsonPath().getString( path: "firstname"));
23     assertTrue( condition: response.jsonPath().getInt( path: "totalprice")>=0);
24 }
25
26 @Test & Shahbaz Ahmad(UST
27 @Order(4)
28 @DisplayName("GET - Fetch booking By ID")
29 void getBookingByIdQueryPath() {
30
31     var response :Response = given() RequestSpecification
32         .spec(commonJsonRequest)
33         .queryParams( parameterName: "id", ...parameterValues: 2)
34         .when()
35         .get( path: "/booking/") Response
36         .then() ValidatableResponse

```

```

@Test & Shahbaz Ahmad(UST
@Order(4)
@DisplayName("GET - Fetch booking By ID")
void getBookingByIdQueryPath() {

    var response :Response = given() RequestSpecification
        .spec(commonJsonRequest)
        .queryParams( parameterName: "id", ...parameterValues: 2)
        .when()
        .get( path: "/booking/") Response
        .then() ValidatableResponse
        .spec(commonJsonResponse)
        .extract() ExtractableResponse<Response>
        .response();

    assertNotNull(response.jsonPath().getString( path: "firstname"));

}

```

```

@Test & Shahbaz Ahmad(UST
@Order(3)
@DisplayName("PUT - update booking")
void updateBookingById() {

    var response :Response = given() RequestSpecification
        .spec(commonJsonRequest)
        .header( headerName: "Cookie", headerValue: "token="+testCreateToken())

```

```

143
144 @Test & Shahbaz Ahmad(UST
145 @Order(3)
146 @DisplayName("PUT - update booking")
147 void updateBookingById() {
148
149     var response : Response = given() RequestSpecification
150         .spec(commonJsonRequest)
151         .header( headerName: "Cookie", headerValue: "token="+testCreateToken())
152         .pathParam( parameterName: "id", parameterValue: 1)
153         .body(requestBody)
154         .when()
155         .put( path: "/booking/{id}") Response
156         .then() ValidatableResponse
157         .spec(commonJsonResponse)
158         .extract() ExtractableResponse<Response>
159         .response();
160
161
162     assertEquals( expected: "Jim", response.jsonPath().getString( path: "firstname"));
163 }
164
165
166
167
168 @Test & Shahbaz Ahmad(UST
169 @Order(5)
170 @DisplayName("DELETE - Delete Booking")
171 void testDeletePet() {
172
173     var response : Response = given() RequestSpecification
174         .spec(commonJsonRequest)
175         .body(requestBody)

```

```

7
8  @Test  & Shahbaz Ahmad(UST
9  @Order(5)
10 @DisplayName("DELETE - Delete Booking")
11 void testDeletePet() {
12
13     var response :Response = given() RequestSpecification
14         .spec(commonJsonRequest)
15         .body(requestBody)
16         .when()
17         .post( path: "/booking") Response
18         .then() ValidatableResponse
19         .spec(commonJsonResponse)
20         .extract() ExtractableResponse<Response>
21         .response();
22     JsonPath json = response.jsonPath();
23     int Id = json.getInt( path: "bookingid");
24
25     given() RequestSpecification
26         .spec(commonJsonRequest)
27         .header( headerName: "Cookie", headerValue: "token="+testCreateToken())
28         .pathParam( parameterName: "id", Id)
29         .when()
30         .delete( path: "/booking/{id}") Response
31         .then() ValidatableResponse
32         .statusCode( expectedStatusCode: 201)
33         .extract() ExtractableResponse<Response>
34         .response();
35
36     var res :Response = given() RequestSpecification
37         .spec(commonJsonRequest)
38         .when()

```

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```

5      given() RequestSpecification
6          .spec(commonJsonRequest)
7          .header( headerName: "Cookie", headerValue: "token="+testCreateToken())
8          .pathParam( parameterName: "id", Id)
9          .when()
10         .delete( path: "/booking/{id}") Response
11         .then() ValidatableResponse
12         .statusCode( expectedStatusCode: 201)
13         .extract() ExtractableResponse<Response>
14         .response();
15
16
17     var res :Response = given() RequestSpecification
18         .spec(commonJsonRequest)
19         .when()
20         .get( path: "/booking/{id}", Id) Response
21         .then() ValidatableResponse
22         .statusCode( expectedStatusCode: 404) // 404
23         .extract().response();
24
25     assertEquals( expected: 404, res.statusCode());
26     assertThat(res.statusCode(), equalTo( operand: 404));
27 }
28
29 }
30
31
32

```

ApiCrudTest Class

- This class performs end-to-end CRUD testing for the Pet API using RestAssured and JUnit 5.
- Uses ordered test execution to follow Create → Read → Update → Delete flow.
- Reuses common request/response specifications for consistency.

setup() (@BeforeAll)

- Initializes the request body before all tests run.
 - Generates a unique pet ID using current timestamp.
 - Sets default values like name and status for test data.
-

tearDown() (@AfterAll)

- Executes after all tests are completed.
- Prints a simple message indicating test completion.
- Can be extended for cleanup or reporting.

testCreateToken

Creates an authentication token using valid admin credentials.

Sends POST request to /auth endpoint with username and password.

Extracts token from response JSON for use in secured requests.

testGetAllBooking

Retrieves all booking records using GET /booking.

Validates that response is JSON and contains booking IDs.

Ensures at least one booking exists in the list.

createBooking

Creates a new booking using POST /booking with request body.

Stores returned booking ID for later test usage.

Verifies booking ID is generated successfully.

getBookingByID

Fetches booking details using a path parameter ID.

Validates response contains customer details like firstname.

Ensures total price returned is valid and non-negative.

getBookingByIdQueryPath

Attempts to fetch booking using query parameter instead of path.

Sends GET request to /booking with query param id.

Validates booking details are returned in response.

updateBookingById

Updates existing booking using PUT /booking/{id}.

Uses authentication token in cookie header for authorization.

Verifies that updated data like firstname is reflected.

testDeleteBooking

Creates a booking and then deletes it using DELETE endpoint.

Uses auth token to authorize deletion request.

Confirms deletion by verifying GET returns 404 status.

Commands

- mvn clean test
- mvn test
- allure generate target/allure-results --clean -o target/allure-report

- allure serve target/allure-results

Execution Result

Metric	Value
--------	-------

Total Tests	6
-------------	---

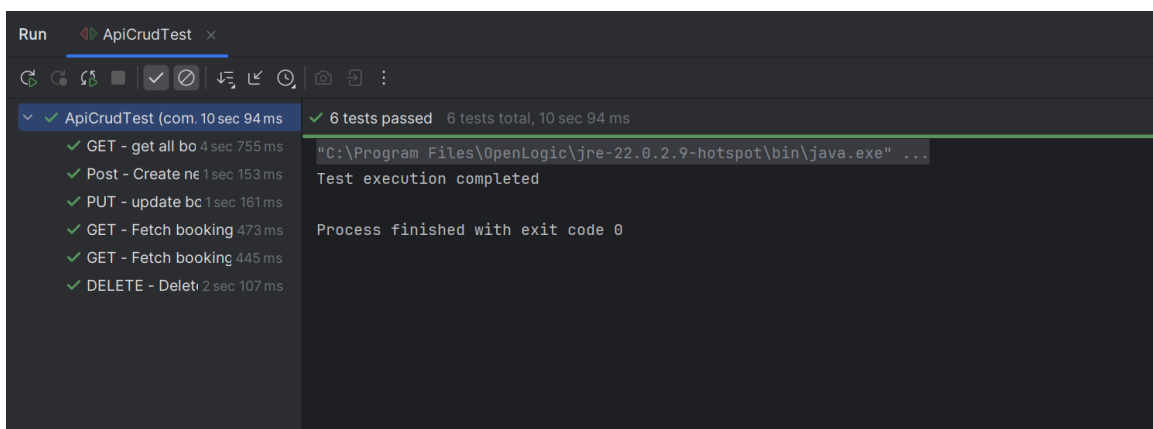
Passed	6
--------	---

Failed	0
--------	---

Skipped	0
---------	---

Execution Status Success

Result Evidence & Screenshots



How to Solve Issues

screenshot

Environment Variables

screenshot

Limitations

Dependent on environment configuration, external API availability, and token expiration.

Conclusion

The API automation framework successfully validates the critical API workflows of the application, ensuring reliability, accuracy, and performance of REST services.

The framework is scalable, maintainable, and follows industry-standard API testing practices. All automated API test scenarios were executed successfully, demonstrating the robustness and stability of the implemented solution.

Key achievements:

- API endpoints validation completed (GET, POST, PUT, DELETE).
- Request and response structure validated using Swagger UI.
- Positive and negative scenarios covered effectively.
- Authentication and secure API access verified.
- Reusability achieved using SpecFactory design pattern.
- Assertions implemented using JUnit and Hamcrest.
- Environment handling managed via .env and TestEnvironment class.
- Allure reporting generated with detailed execution insights.

- Test execution completed successfully with expected results.